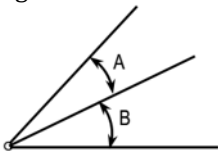


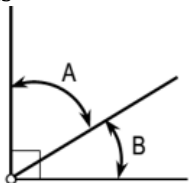


Other Types of Angles

1. Adjacent angles – two angles with a common leg

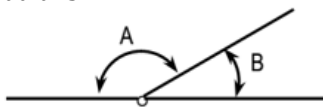


2. Complementary angles – two angles whose sum is a right angle or 90° or $\pi/2$ radians



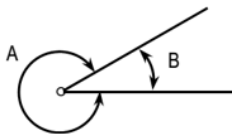
Angle A and angle B are complementary angles. Angle A is the complement of angle B and vice versa.

3. Supplementary angles – two angles whose sum is a straight angle or 180° or π radians



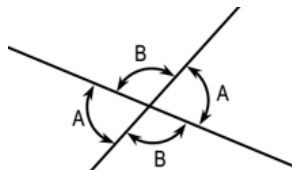
Angle A and angle B are supplementary angles. Angle A is the supplement of angle B and vice versa.

4. Explementary angles – two angles whose sum is a perigon or 360° or 2π radians



Angle A and angle B are Explementary angles. Angle A is the explement of angle B and vice versa.

5. Vertical angles – angles formed by two intersecting lines. Vertical angles are equal.



Angle A and angle B are vertical angles. These angles are called vertical angles because each side of one is an extension through the vertex of a side of the other.

Different units of angles

- Degree** – The degree is a non-SI unit approved in 1969 by the General Conference on Weights and Measures. This unit was established by the Babylonians more than 4000 years ago. A degree is divided into 60 minutes; a minute is divided into 60 seconds.
- Radian** – The radian is the standard angular measure in the International System of Units (SI). A radian is the angle between two radii of a circle which cut off on the circumference an arc equal in length to the radius.
- Gon** – The gon or centesimal degree is $1/400$ th of the full circle. A right angle is 100 gons. A gon is divided into 100 centesimal minutes; one centesimal minutes is divided into 100 centesimal seconds. This unit is also known as grad or grade and is used in surveying works and aircraft navigation.
- Mil** – The mil is a measure of an angle which is $1/6400$ of the full circle. A right angle is 1600 mils.

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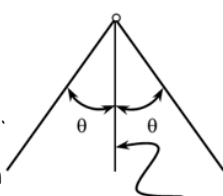


For easy conversion from one unit to another, just remember the measure of an angle with corresponding value in one revolution. Then apply ratio and proportion!

Unit	One revolution
Degree	360
Radian	2π
Gon	400
Mil	6400

Angle Bisector

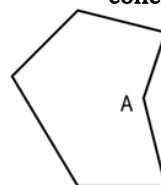
A bisector is a straight line which divides a geometric figure into two equal figures. An angle bisector is a line which divides an angle into two equal angles



Angle bise



Convex polygon



Concave polygon

Polygon

A plane figure with three or more angles. It has as many sides as angles. The sides of the polygon are straight lines. The term "polygon" comes from Greek words "poly" which means "many" and "gonia" which means "angles".
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Regular polygons

A regular polygon is a polygon having all sides equal and all angles equal. Table below shows the regular polygons:

Name	Sides	Angle
Equilateral triangle	3	60°
Square	4	90°
Regular pentagon	5	108°
Regular hexagon	6	120°
Regular heptagon	7	$\approx 128.57^\circ$
Regular octagon	8	135°
Regular nonagon	9	140°
Regular decagon	10	144°
Regular hectagon	100	176.4°
Regular megagon	10^6	179.99964°
Regular googolgon	100^{100}	$\approx 180^\circ$

Names of polygons

Polygons are named according to the number of sides or vertices. Below are the polygons with their corresponding names.

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Number of sides	Name
3	Triangle
4	Quadrilateral or Tetragon
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon
11	Undecagon
12	Dodecagon
100	Hectagon
1000000	Megagon
10^{100}	Googolgon
n	n-gon

Convex polygon is one having each interior angle less than 180° while **concave polygon** is one having an interior angle greater than 180° .



Did you know?

Since 2013 (under new examiners) to date, Excel produced more ECE topnotchers than the total topnotchers of all other review centers combined!